

IGI Coursework 2 – Medical Scene Understanding

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1 Task 1

1.1 (i)

Let the (right, up, view) axes of each camera be (x, y, z) , respectively. Each of the epipolar lines is a horizontal line with equation: $y = 0$.

1.2 (ii)

The equation of the epipolar plane is as follows.

$$-z_i y + y_i z = 0 \quad (1)$$

1.3 (iii)

Equation of line in 3d in world coordinates is as follows.

$$r_{3d_world}(t) = \begin{bmatrix} -5 - tz_i \\ \frac{7y_i}{z_i} \\ 7 \end{bmatrix} \quad (2)$$

Equation of line in 3d in left camera coordinate frame is as follows.

$$r_{3d_left_camera}(t) = \begin{bmatrix} -tz_i \\ \frac{7y_i}{z_i} \\ 7 \end{bmatrix} \quad (3)$$

The 2d line equation in the image plane of the left camera is impossible to calculate as the intrinsic parameters of the left camera are not known.

1.4 (iv)

Use the patient's belly button as the feature point. Use shape from stereo to estimate the height of the belly button over time. Extract the peaks of the time-series and divide over the time period. This is the respiratory rate. The shape from stereo can be implemented similarly as in [task 3 \(e\)](#).

2 Task 2

2.1 (i)

$$\begin{bmatrix} R & | & T \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & | & 0 \\ 0 & 0 & 1 & | & -d \\ 1 & 0 & 0 & | & 6d \end{bmatrix} \quad (4)$$

2.2 (ii)

$$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 350 \\ 130 \end{bmatrix} \quad (5)$$

3 Task 3

3.1 (a)

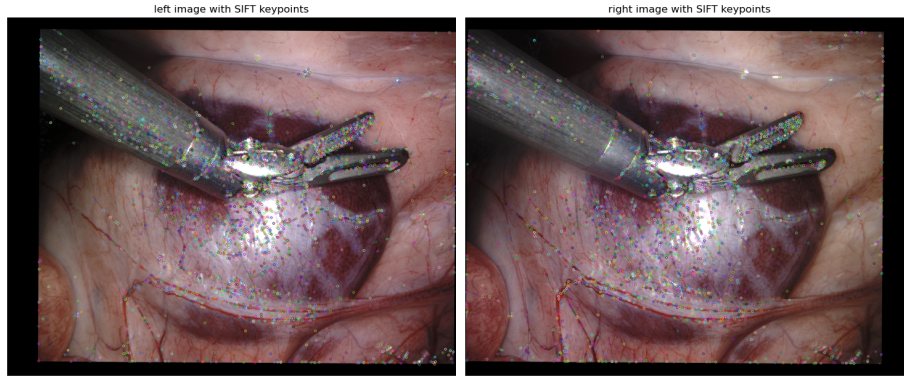


Figure 1: SIFT features

First, I do rectification. I use the SIFT feature-detection algorithm for several reasons. It is strong at detecting features in complex and textured environments (such as a human's internal anatomy). It is also robust to scale, rotation and illumination changes. SIFT is computationally expensive but this is not an issue as this is not a real-time processing scenario. I use the OpenCV (cv2) Python library with the following functions: `SIFT_create`, `detectAndCompute`, `drawKeypoints`.

3.2 (b)

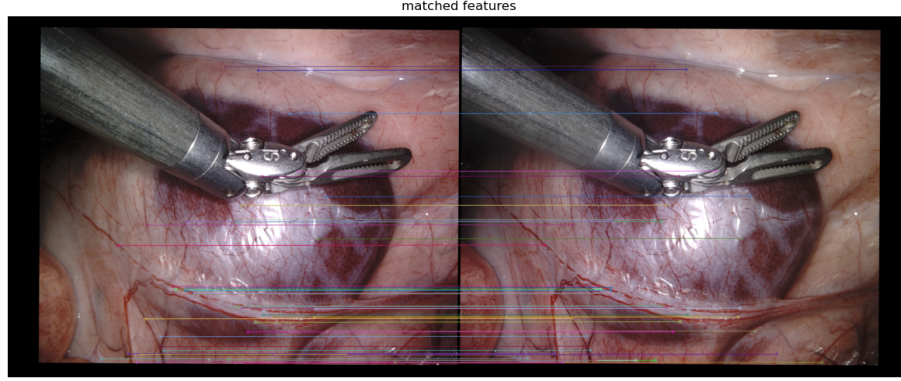


Figure 2: Matched SIFT features

I use brute-force matching. I perform a ratio test on the top 2 matches with $ratio = 0.75$. I sort the descriptor pairs by distance ascending and display the first 50 pairs.

3.3 (c)

Fundamental matrix F_1 obtained from stereo matching is shown **below**. To calculate it I convert the matched descriptor pairs to matched coordinate pairs. I then run OpenCV's `findFundamentalMat` function using the `FM_RANSAC` flag. I use the RANSAC algorithm due to its robustness to outliers and low signal-to-noise ratio scenarios.

$$F_1 = \begin{bmatrix} -6.22 \cdot 10^{-8} & 8.33 \cdot 10^{-6} & 3.23 \cdot 10^{-3} \\ -7.82 \cdot 10^{-6} & 8.97 \cdot 10^{-7} & -8.64 \cdot 10^{-2} \\ -3.52 \cdot 10^{-3} & 8.51 \cdot 10^{-2} & 1.00 \cdot 10^0 \end{bmatrix} \quad (6)$$

The fundamental matrix F_2 obtained from the stereo camera parameters is presented **below**. I use the formula introduced in the lecture, which follows.

$$F = [K't]_x K' R K^{-1} \quad (7)$$

$$F_2 = \begin{bmatrix} 2.19 \cdot 10^{-8} & -6.26 \cdot 10^{-7} & -1.84 \cdot 10^{-2} \\ 3.98 \cdot 10^{-6} & -1.19 \cdot 10^{-7} & -2.87 \cdot 10^0 \\ 1.67 \cdot 10^{-2} & 2.87 \cdot 10^0 & 1.00 \cdot 10^0 \end{bmatrix} \quad (8)$$

The maximum absolute element-wise ratio between F_1 and F_2 is presented **below**.

$$ratio = \begin{bmatrix} 3 & 13 & 6 \\ 2 & 8 & 33 \\ 5 & 34 & 1 \end{bmatrix} \quad (9)$$

F_1 and F_2 produce slightly different results. The reason for this is that each of them is calculated differently and each calculation involves some potential errors. Feature detection and matching used for F_1 are prone to errors due to lack of texture and repetitive patterns. The output of RANSAC depends on the inlier threshold and the number of iterations. Modifying these parameters will result in different fundamental matrices. F_2 is obtained using camera parameters. The latter are calculated based on a calibration process. Some common errors here include: insufficient calibration data, improper placement of the calibration pattern or changes in lighting conditions between calibration shots. Additionally, the 5-parameter distortion model used is only an imperfect approximation of the lens distortion. There exist alternative distortion models with a higher number of parameters. Finally, the rectification process might introduce resampling artifacts, e.g. blurring, aliasing or loss of detail. All these errors contribute to slightly different fundamental matrices.

F_2 is usually more accurate. This is due to the fact that camera calibration is less error-prone than feature extraction and matching. Another argument for this is the visualisation of **epipolar lines calculated using F_1 and those calculated using F_2** . The former are more skew, while the latter are more horizontal. Since the images are rectified, it is expected that the all epipolar lines should be horizontal.

3.4 (d)

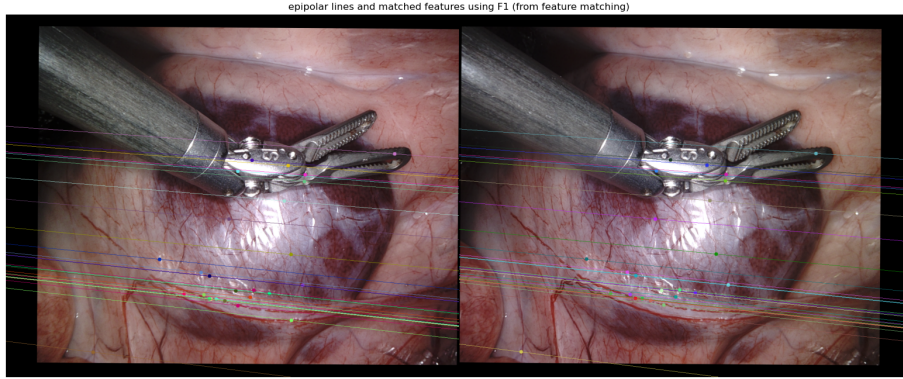


Figure 3: Epipolar lines obtained using F_1

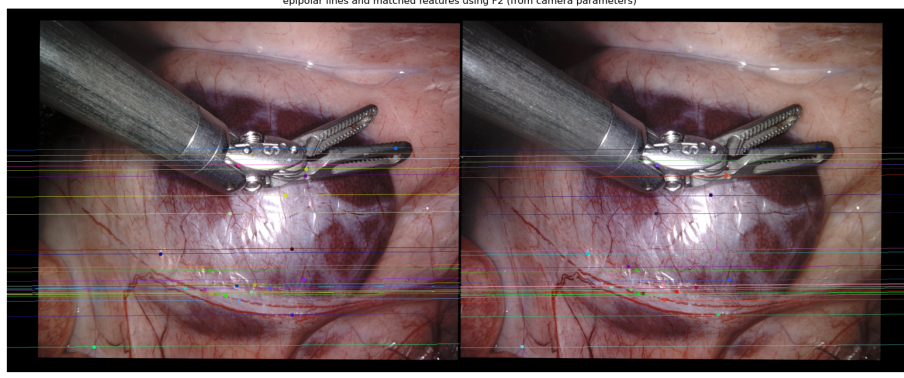


Figure 4: Epipolar lines obtained using F_2

The matches satisfy the epipolar constraints iff

$$x'^T F_2 x < 10^{-3} \quad (10)$$

. This condition is calculated for all matches in the list. A boolean mask is calculated and applied to the original set of matches. This yields the set of matches that satisfy the epipolar constraint. It can be seen that F_2 produces better results (epipolar lines are more horizontal) than F_1 .

3.5 (e)

The diameter of the shaft is $7.40mm$. The disparity map is calculated (shown [below](#)). The method `cv2.stereoRectify` is used to obtain the reprojection matrix Q . Calling `cv2.reprojectImageTo3D(disparity, Q)` yields the 3D coordinates of each point in the image. The 2D pixel coordinates of 9 approximately evenly spaced points are manually obtained from the visible circle on the shaft in using the left image (as shown in the highlight in the [figure below](#)). The 3D coordinates of these 8 points are then looked up. A 3D circle is fitted to the data points. First, a 3D plane is fitted to the data. Later, RANSAC is used - 3 data points are sampled in each iteration. The cost function is as follows:

$$\sum (d_{\text{centre}} - r)^2 + \sum d_{\text{plane}}^6 \quad (11)$$

, where d_{centre} is the distance from the data point to the circle centre, r is the circle radius and d_{plane} is the distance from the data point to the 3D plane. The center, radius and normal of the 3d circle are computed (points sampled from computed circle and data points are visualised in [Fig. 7](#) and [Fig. 8](#)).

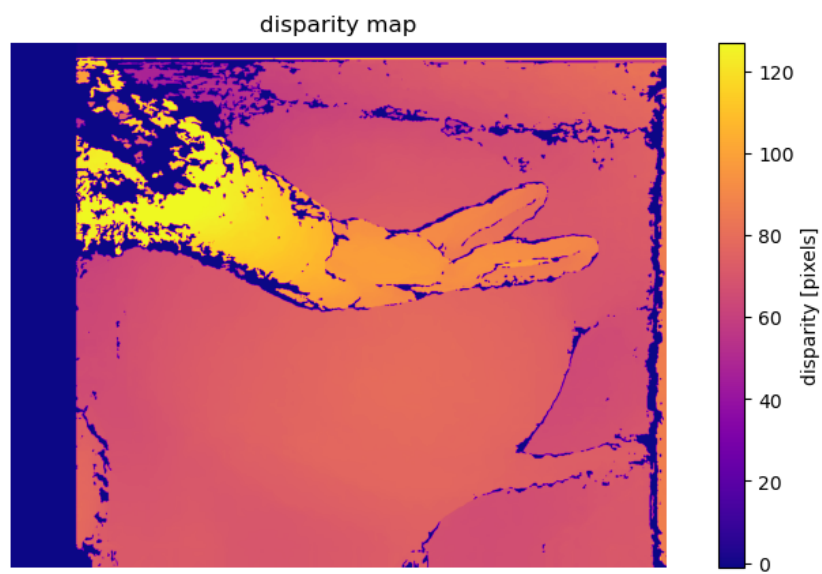


Figure 5: Disparity map

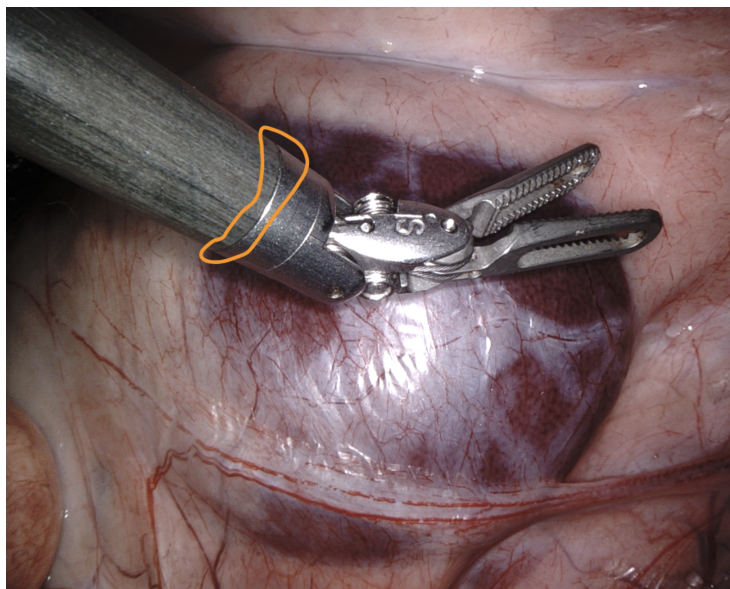


Figure 6: Region used for pixel sampling for 3d reconstruction

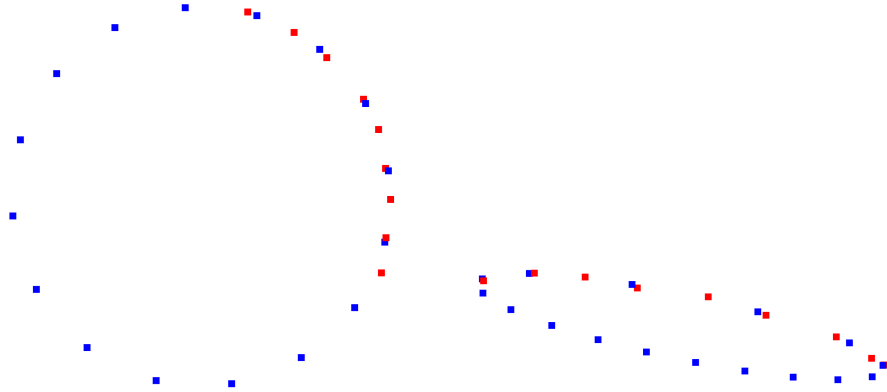


Figure 7: 3D plot of the sampled data points (red) and reconstructed shaft circumference (blue) - top view

Figure 8: 3D plot of the sampled data points (red) and reconstructed shaft circumference (blue) - side view

4 Task 4

4.1 (a)

$$tf_0(V) = \begin{cases} 255 & \text{if } 10 < V < 25 \vee V > 80 \\ 167 & \text{otherwise} \end{cases} \quad (12)$$

$$tf_1(V) = \begin{cases} 255 & \text{if } 10 < V < 25 \vee V > 80 \\ 0 & \text{otherwise} \end{cases} \quad (13)$$

$$tf_2(V) = \begin{cases} 255 & \text{if } 10 < V < 25 \vee V > 80 \\ 0 & \text{otherwise} \end{cases} \quad (14)$$

$$tf_3(V) = \begin{cases} 0.5 & \text{if } 10 < V < 25 \\ 1.0 & \text{otherwise} \end{cases} \quad (15)$$

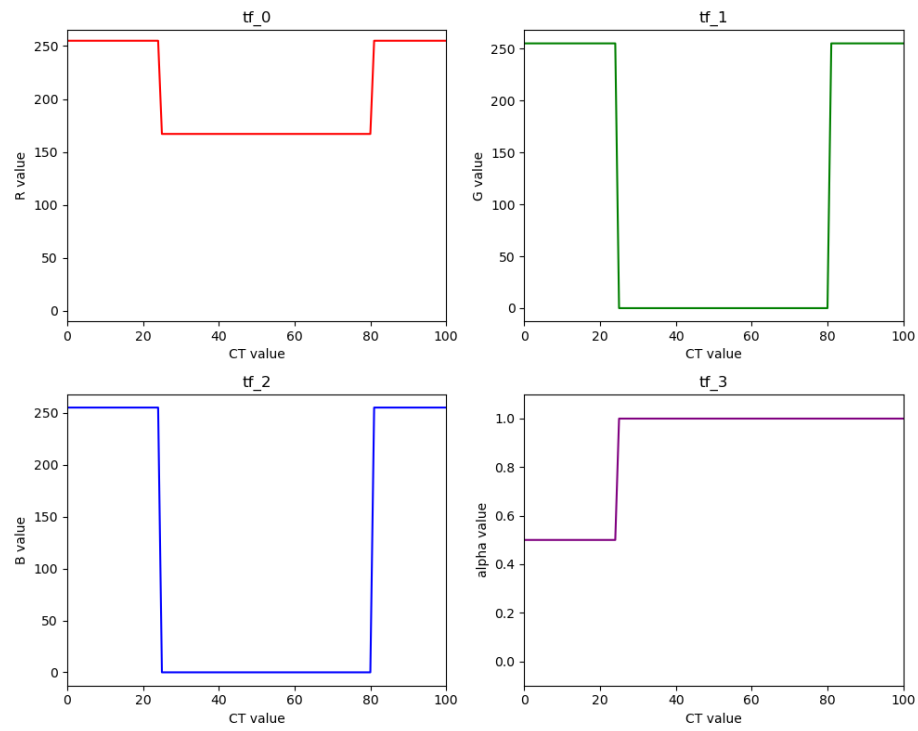


Figure 9: Transfer functions

4.2 (b)

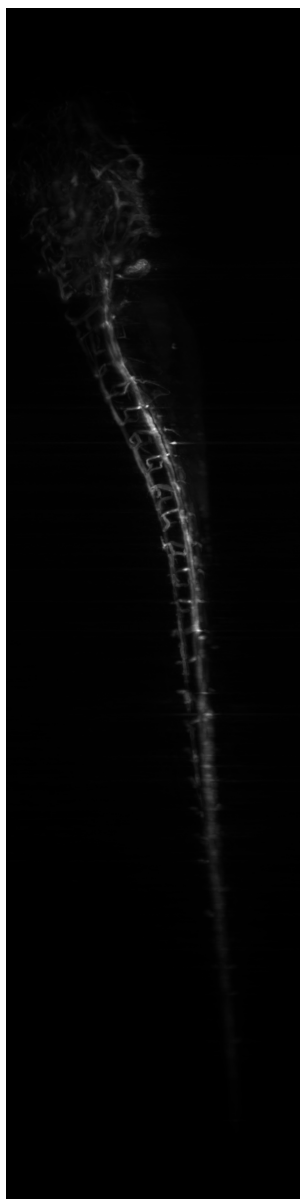


Figure 10: X-axis projection

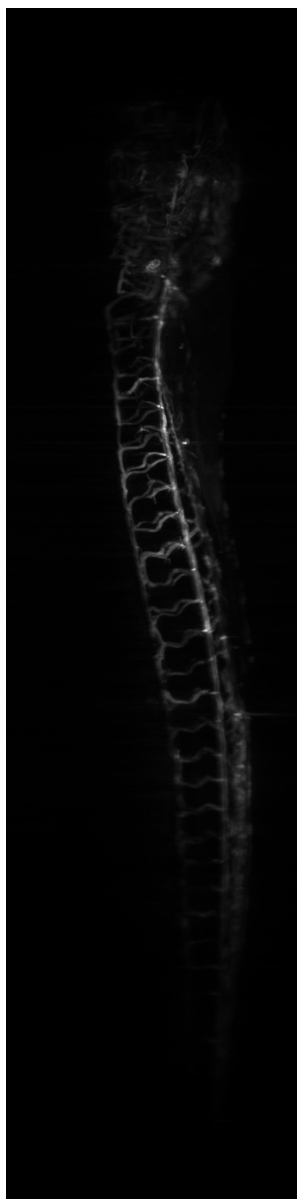


Figure 11: Y-axis projection

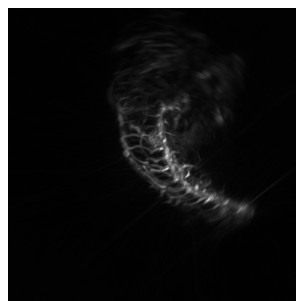


Figure 12: Z-axis projection